

Capstone Project Monthly Progress Report

Month: March 2026

Project Title: Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection

Project ID: CAPSTONE_2022_24250_3

Student Name:: D.Satya Koushik

Student Roll Number: AP22110010454

Total Number of Students in Team: 04

Details of Team Members:

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Y. Teja (Roll No. AP22110010409)

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Supervisor: Dr.Dilip Kumar

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1 Project Overview

This research work, entitled “**Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection**,” offers a safe method to detect fingerprint template reconstruction attacks prior to the attack’s success at compromising biometric authentication systems. The method combines a ResNet-50 deep learning model for robust feature extraction with a hybrid dynamic projection technique to produce secure biometric templates. Based on the analysis of projection consistency, correlation, and statistical differences between original and reconstructed templates, the system is able to successfully detect malicious attacks while maintaining the efficiency of biometric authentication. The system is tested using the CASIA-FingerprintV5 dataset

2 Work Completed During Last 1 Month

- Developed an effective secure projection-based fingerprint template protection framework that identifies template reconstruction attacks.
- Improved ResNet-50 architecture to achieve an effective 512-dimensional fingerprint embedding with strong feature extraction capabilities.
- Created an effective fingerprint detection system using projection consistency, including statistical analysis, correlation analysis, and distribution analysis.

- Performed extensive testing of the system using the CASIA-FingerprintV5 dataset with various training and testing scenarios.
- Monitored the performance of the system using critical parameters such as Accuracy, Precision, Recall, F1-score, AUC-ROC, and equal error rate.
- Generated visualizations of results using confusion matrix visualization, ROC curves, and training graphs Updated all experimental results in the final research report.

3 Work Planned for this Month

- Extend analysis to cross-dataset and cross-sensor scenarios for robustness evaluation.
- Develop and analyze advanced reconstruction attack strategies.
- Optimize the framework to reduce computational complexity.
- Compare the proposed method with existing techniques in the literature.
- Fine-tune detection thresholds to achieve optimal FPR and high GAR.
- Enhance experimental evaluation using statistical validation methods.
- Improve result visualization, and complete final testing, validation, and documentation.

4 Progress Status

- Overall completion: 100%
- The proposed framework was successfully designed and implemented.
- Integration of ResNet-50 model was successfully achieved.
- Gained hands-on experience in deep learning-based biometric authentication using ResNet-50.
- Developed understanding of template security, reconstruction attacks, and detection methods.
- Implemented projection-based protection techniques and machine learning models for detection.
- Enhanced skills in model training, hyperparameter tuning, and performance evaluation using metrics like Accuracy, F1-score, and AUC-ROC.

5 Challenges/Issues Faced

- Difficulty in managing and preprocessing the fingerprint data set of large scale.
- The High computational requirements for training the ResNet50 model.

- The task of creating realistic samples for reconstruction attack generation, whether algorithmic or learning-based, without leaking data from the training process to the testing process was a technical challenge.
- Fine-tuning the detection threshold to achieve optimal FPR and GAR rates. Unavailability of diverse data sets for cross-data set validation.
- Overfitting of the model and achieving generalization.
- Time complexity involved in optimizing the system for real-time processing. Interpreting the statistical results and validating the detection approach.
- Integration of all the components into a single system.

6 Learning Outcomes

- Acquired hands-on experience with deep learning-based biometric authentication using the ResNet-50 model for efficient fingerprint feature extraction.
- Achieved a deeper understanding of biometric template security techniques, template reconstruction attacks, and detection methods.
- Gained practical experience in the implementation of projection-based template protection and consistency analysis methods.
- Learned to develop and test machine learning classifiers (Logistic Regression, MLP, SVM) for attack detection problems.
- Enhanced skills in model training, hyperparameter optimization, and performance optimization using PyTorch.
- Gained experience in the application of evaluation metrics such as Accuracy, Precision, Recall, F1-score, AUC-ROC, and Equal Error Rate (EER) for security-oriented systems.

7 Plan for Next Month

- Submit the research paper to a conference/journal and prepare for presentation.
- Extend experiments to cross-dataset and cross-sensor evaluations.
- Implement more advanced reconstruction attack scenarios. Optimize the model for faster and real-time performance.
- Submit the research paper to a conference/journal and prepare presentation materials (PPT/poster).

Expected deliverables

- Conference-ready research paper submission.
- Optimized detection system ready for deployment-level testing.

- Presentation slides/poster
- Experimental results for extended analysis

8 Student Individual Contribution

- **K.Babnadh**

Led the design and implementation of the framework for fingerprint template reconstruction attack detection, integrated ResNet-50 for feature extraction, handled model training and evaluation, and contributed to analysis, paper writing, and team coordination for timely completion.

- **Y.Teja**

Worked on preprocessing the CASIA-FingerprintV5 dataset, assisted in feature extraction and model evaluation, contributed to testing and validation, improved data quality, and supported documentation refinement.

- **M.Nithin Sai**

Developed and implemented the hybrid dynamic projection technique, optimized the framework for efficiency, contributed to algorithm design and parameter tuning, and strengthened the methodology section.

- **D.Satya Koushik**

Handled system testing, validation, and comparison with existing methods, analyzed results with visualizations, contributed to documentation and presentation preparation, and supported final review for clarity and consistency.

9. Supervisor's Observation

Student Completed: 100% of the planned/assigned task during the last month

Comments:

Number of times students meet with supervisor (Online or Offline): offline

Satisfied with the Student's Work: Satisfied Not Satisfied

Supervisor Signature: _____

Date: DD/MM/YY